ACTUA



PRODUCT VERSION

This technical data sheet is related to following product versions of ECU Actua:

	ECU ACTUA OTS1		ECU ACTUA FULL	
PN#	6200195/1x		6200156/6x	
Power supply	12 Vdc	24 Vdc	12 Vdc	24 Vdc
Contents				
Inputs / outputs	X	X	X	X
CAN BUS 1, 2, 3, 4	X	X	X	X
LIN BUS 1, 2	X	X	X	X
RTC	X	X	X	X
Sensor supply 1, 2	X	X	X	X
H-Bridge	X	n.a.	X	n.a.
Additional contents				
Eth-BR			X	X
USB			X	X
RS232			X	X
Accelerometer			X	X
External flash			X	X
Sensor supply 15 V			X	X
Certifications				
Type approval ECE ONU reg10-05 E3	X	X	X[2]	X[2]
MTA Studio ID[1]	0204	0304	0004	0104

GENERAL DESCRIPTION

Actua is a universal controller designed for agricultural or off-highway vehicles, on wheels or tracks.
Power supply is 12 and 24 Vdc.

ECU Actua common features:
- 48 configurable inputs;
- 8 frequency inputs;
- 2 fixed analog inputs;
- 32 power outputs;
- 1 H-Bridge driver (only for 12 Vdc applications);
- 2 auxiliary power supplies (5 Vdc or 8 Vdc);
- 6 communication lines (4 CAN, 2 LIN).

ECU Actua Full (PN# 6200156/6x) includes the following options, not mounted on the ECU Actua OTS1:
- 3 communication lines:
  - USB: device mode available for service purposes[3] only;
  - Ethernet BroadR-Reach;
  - RS-232 not available as vehicle communication bus;
- 1 accelerometer;
- 1 serial fash memory;
- 1 auxiliary power supply (15 Vdc for 24 Vdc applications).

Actua can be used as a controller for:
- drive relays, solenoids or other resistive loads;
- analog signal interface and monitor;
- digital and frequency signals from external sensors.

An RTC module is available to give timing information, without backup battery.

It features 2 memories for permanent data storage (EEPROM and an optional serial flash).

Actua is configurable with MTA Studio and MTA Gate[4].

MTA STUDIO

MTA Studio is a specific tool used to program dashboards, displays and electronic control units which have been developed with customizable functions allowing the customers to create “tailored” devices.

MTA GATE

MTA Gate is a specific tool designed for EOL department and/or for technical service points use.

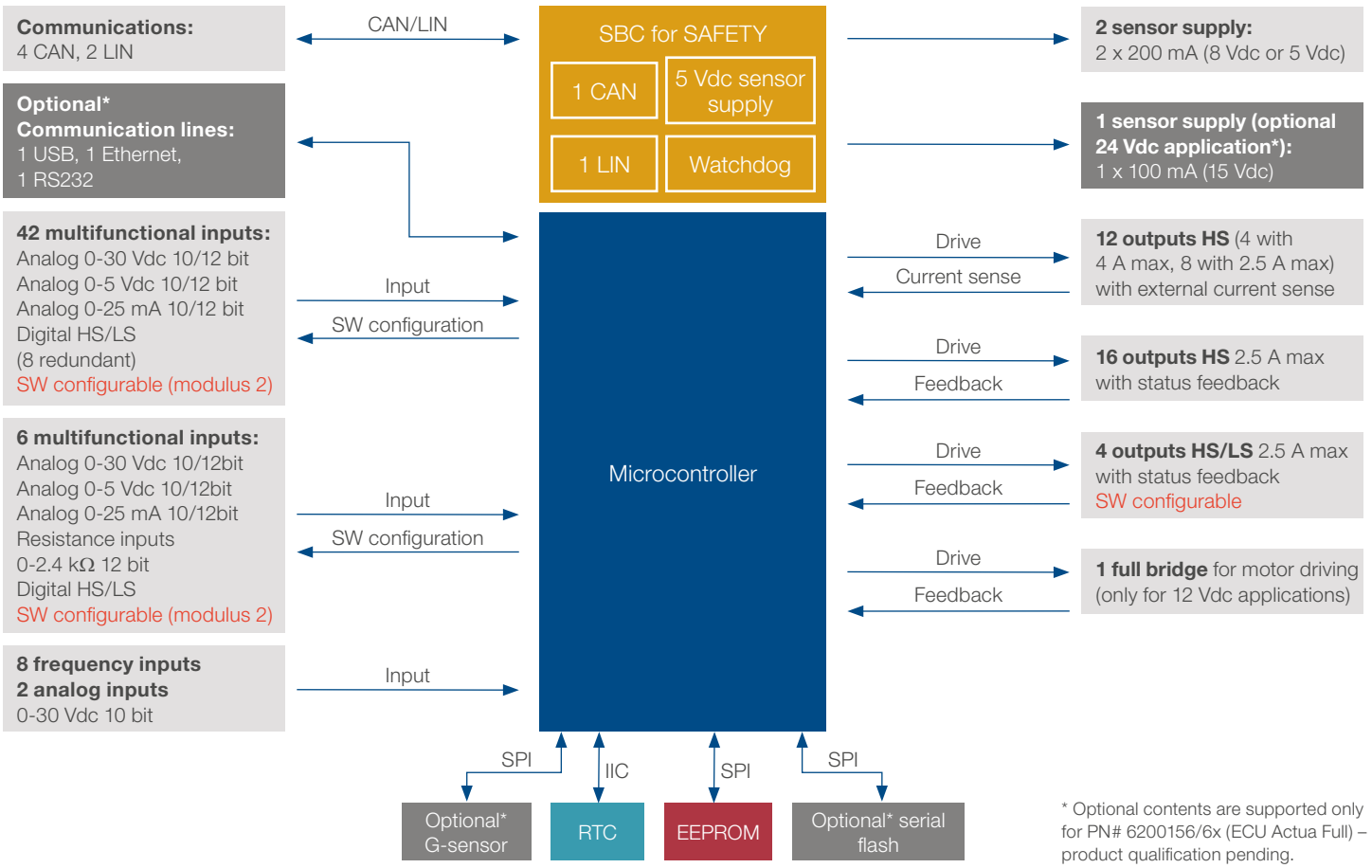
1. The “MTA Studio ID” here listed are referred to the MTA Studio variants, for ECU Actua product family, written on the product label. Please contact MTA for any info about it.

2. Extension of ECE ONU reg. 10-05 to the ECU Actua Full is not already performed because the ECU Actua Full product qualification is pending; it will be done only after the product qualification. With the nominal load connected, the low side outputs are activated without risk of damaging any part, since the current is limited by the load itself.

3. “Feature available for service” means that:
- the user can use this peripheral only for service purposes, not in standard operational conditions; e.g.: software diagnostic conditions on vehicle;
- the product qualification is limited to the verification of peripheral, not to guarantee it in all standard product working conditions;
- the previous limitations must be clearly identified on the technical data sheet;
- the main limitation is due to the lack of customer’s specifications about his needs. For this reason customer’s requests about specific use case may imply an additional R&D software development and product qualification to be evaluated case by case.

4. Please contact MTA to verify which features can be programmed with MTA Studio and MTA Gate. Custom development could be evaluated by MTA in case of specific functionalities needed and not already available.

BLOCK DIAGRAM



* Optional contents are supported only for PN# 6200156/6x (ECU Actua Full) – product qualification pending.

SPECIFICATIONS

CHARACTERISTICS	DESCRIPTION	QTY
Operating voltage		
Nominal	12 Vdc and 24 Vdc	
Supply voltage, permissible range	9÷16 Vdc (full functionality) 18÷32 Vdc (full functionality)	
Operating temperature		
External ambient temperature	-40 to +85 °C	
Storage temperature		
External ambient temperature	-40 to +85 °C	
Protection rate		
IP67		
Weight		
1450 g		
Mounting location (ref. TS 50_16)		
Location B (outside cab/ chassis mounted)		
Microcontroller		
PowerPC	NXP SPC5746C family	
Clock frequency		
MHz	160 MHz + 80 MHz (dual core)	

Internal memory capacity in microcontroller	
Flash	3 MB
RAM	512 KB
EEPROM	
Bus	SPI
Size	32 Kbit
External flash memory (optional)	
Bus	SPI
Size	512 Mbit
Accelerometer (optional)	
Bus	SPI
Axis	3
Range (SW configurable)	±2 g or ±6 g
Real time clock	
Non battery powered	Clock time deviation: ±2.17 sec per day (-40÷+85 °C)
Temperature sensor	
Internal	✓
Battery voltage measurement	
Internal	Accuracy: ±10% of read value

CHARACTERISTICS	DESCRIPTION	QTY
Interfaces		**9**
CAN HS (up to 1Mb/s)	ISO 11898	4
LIN	LIN 2.1	2
RS-232 (optional)	Up to 115 Kbaud	1
BroadR-Reach Ethernet (optional)	Up to 100 Mb/s	1
USB (optional)	Device	1
CAN bus termination resistors		
Termination resistor	Fixed terminator resistors externally required	
Current consumption		
Without load, 12 Vdc system	< 250 mA at key on < 1 mA at key off (8÷16 Vdc) < 5 mA at key off (6÷8 Vdc)	
Without load, 24 Vdc system	< 350 mA at key on < 1mA at key off (18÷28 Vdc) < 4mA at key off (28÷32 Vdc)	
Fuses		
External in supply path	To be dimensioned for a maximum continuous current consumption of 32 A	
Constant voltage source		**3**
200 mA	5 Vdc ±4% (FS), 8 Vdc ±4%(FS)	2
100 mA for 24 Vdc only (optional)	15 Vdc ±4% (FS)	1
Analog inputs (configurable in pairs)		**40**
0-30 Vdc	12 bit	up to 40
0-5 Vdc	12 bit	up to 40
0-25 mA	12 bit	up to 40
Resistance inputs (configurable as low side type)		**6**
0-2.4 kΩ (configurable in pairs)	12 bit	up to 6
Digital inputs (configurable in pairs)		**40**
With pull-up to battery	10 kΩ	up to 34
With pull-up to 5 Vdc	2.2 kΩ	up to 6
With pull-down	60.2 kΩ	up to 40
Redundant analogue inputs (configurable in pairs)		**8**
0-30 Vdc	10 bit and 12 bit (except IN_43: 10 bit both)	up to 8
0-5 Vdc	10bit and 12bit (except IN_43: 10 bit both)	up to 8
0-25 mA	10bit and 12bit (except IN_43: 10 bit both)	up to 8
Redundant digital inputs (configurable in pairs)		**8**
Digital input with pull-up to battery	10 kΩ	up to 8
Digital input with pull-down	60.2 kΩ	up to 8

Spare analogue inputs (fixed pull-down)		**2**
Input resistance	6.9 kΩ	
0-30 Vdc	10 bit	2
Frequency inputs (fixed pull-up)		**6**
Active inputs	Up to 5 kHz	up to 6
With pull-up to battery	Up to 5 kHz, 10 kΩ	up to 4
With pull-up to battery	Up to 5 kHz, 62 kΩ	up to 2
Frequency inputs (without pull-up)		**2**
Active inputs	Up to 5 kHz	up to 2
High side outputs (not configurable)		**28**
Current controlled PWM output	Max. 4 A, PWM up to 250 Hz	4
No current controlled PWM output	Max. 2.5 A, PWM up to 250 Hz	8
ON/OFF HSDs	Max. 2.5 A, no PWM	16
High side/low side outputs (configurable)		**4**
No current controlled PWM output (high side)	Max. 2.5 A, PWM up to 250 Hz	up to 4
No current controlled PWM output (low side)	Max. 2.5 A, PWM up to 250 Hz	up to 4
H-Bridge (only for 12Vdc applications)		**1**
Peak current	15 A	
RMS Current	3 A	
Fault detection in the event of cable break and short circuit		
Inputs	n.a.	
Outputs	✓	
Reverse polarity protection		
Power supply	✓	
Inputs	✓	
Outputs	HSD outputs ✓ LSD outputs: it is mandatory to use a fuse sized 2.5 A maximum, connected in series to the pin connector, to protect the related Actua output	
Electrostatic discharge ESD		
Handling and servicing	±6 kV	
Operational conditions	±15 kV	
Wake-up		
From key, CAN1, CAN4, LIN1, LIN2, ETH (optional)	CAN 4 always active	
Materials and other mechanical characteristics		
Aluminium	EN AC 46100	
Surface treatment	Electro paint black QT 32-9578 / Stazzi product BASF or equivalent	

CHARACTERISTICS

MULTIFUNCTIONAL INPUTS

Inputs can be used as:

- digital input from 0 Vdc to Vbat (50 of 56) (pins: 9, 10, 12, 13, 14, 15, 17, 18, 19, 23, 24, 25, 26, 30, 31, 32, 35, 36, 37, 39, 40, 41, 45, 46, 52, 53, 54, 55, 56, 57, 58, 60, 61, 67, 70, 75, 76, 77, 79, 80, 101, 102, 103, 104, 106, 107, 108, 115, 118, 119):
  - logic level 0 when voltage input ≤ 1.68 Vdc;
  - logic level 1 when voltage input ≥ 3.67 Vdc;
- digital input from 0 Vdc to 5 Vdc (6 of 56) (pins: 8, 34, 78, 121, 122, 134):
  - logic level 0 when voltage input ≤ 1.68 Vdc;
  - logic level 1 when voltage input ≥ 3.67 Vdc;
- frequency input open collector from 0 Vdc to Vbat and maximum frequency 5 kHz (accuracy: ±5%) (8 of 8) (pins: 101, 102, 103, 104, 106, 107, 108, 115):
  - logic level 0 when voltage input ≤ 1.12 Vdc;
  - logic level 1 when voltage input ≥ 2.45 Vdc;
  - maximum frequency 5 kHz with push-pull type external sensor;
  - maximum frequency 5 kHz with NPN open collector / N open drain type external sensor;
  - it is not possible to detect duty cycle of frequency signals;
  - IN_49 & IN_50 (pins: 101, 102) have 62 kΩ fixed pull-up to battery;
  - IN_51 & IN_52 (pins: 103, 104) are without pull-up;
  - IN_53 to IN_56 (pins: 106, 107, 108, 115) have 10 kΩ fixed pull-up to battery.
- analog input (48 of 50) (pins: 8, 9, 10, 12, 13, 14, 15, 17, 18, 19, 23, 24, 25, 26, 30, 31, 32, 34, 35, 36, 37, 39, 40, 41, 45, 46, 52, 53, 54, 55, 56, 57, 58, 60, 61, 67, 70, 75, 76, 77, 78, 79, 80, 118, 119, 121, 122, 134) with range:
  - 0-5 Vdc (accuracy: ±10% in the range 0.5-5 Vdc; ±20% for values < 0.5 Vdc);
  - 0-30 Vdc (accuracy: ±10% in the range 5-30 Vdc; ±20% in the range 3÷5 Vdc)[5];
  - 0-25 mA (accuracy: ±10% in the range 4-25 mA; ±20% for values < 4 mA).
- analog input (2 of 50) (pins: 132, 133) with range:
  - 0-30 Vdc (accuracy: ±10% in the range 5-30 Vdc; ±20% in the range 3÷5 Vdc)[6];
- resistive input (6 of 50) (pins: 8, 34, 78, 121, 122, 134) with range:
  - 0-2.4 kΩ (accuracy: ±10% of read values in the range 100 Ω-2.4 kΩ; ±20% of read values in the range 30 Ω-100 Ω)[7].

For each input the presence of a pull-up or a pull-down can be configurated to interface with most sensor types, except frequency inputs (six of eight have pull-up only and two of eight are without pull-up/pull-down).

Configuration of 48 multifunctional inputs is available in pairs as listed below:
(IN_1 & IN_2), (IN_3 & IN_4), …, (IN_47 & IN_48)

Actua is provided with a dedicated input for key (terminal 15).

Wake-up functionality

- 1 key dedicated input provides wake up capability (pin: 20).
- 2 CAN with wake-up capability (CAN1 software configurable; CAN 4 always active).
- 2 LIN with wake-up capability (LIN1 and LIN2).
- 1 BroadR-Reach with Ethernet wake-up capability (optional).

OUTPUTS

Outputs (16 of 32)[8] can be driven in PWM mode with frequency configurable via software in the range 0 to 250 Hz (common value for all outputs) and duty cycle in the range 5÷100% (resolution: 1%).
Outputs are monitored and protected.
Outputs status is available in a periodic dedicated CAN message that provides this information:

- output is ok;
- output in short circuit, to ground or battery or with a high current consumption over a threshold (only for drivers with current sense);
- output in open circuit.

Maximum current for each output is 4 A or 2.5 A, as specified in the pinout table.

Output load characteristics

Outputs 4 A:

- minimum load resistance: 3.5 Ω @ 14 Vdc; 4 Ω @ 16 Vdc; 7 Ω @ 28 Vdc; 8 Ω @ 32 Vdc;
- maximum load inductance: 500 mH @ 4 A;
- for inductive loads with characteristics out of these requirements a dedicated evaluation shall be done.
- current sense accuracy:
  - without calibration procedure (default): ±35% in range (0.5÷1) A, ±30% in range (1÷2) A, ±20% in range (2÷4) A;
  - with calibration procedure (custom solution[9]): ±13% in range (0.5÷4) A.

Outputs 2.5A:

- minimum load resistance: 5.6 Ω @ 14 Vdc; 6.4 Ω @ 16 Vdc; 11.2 Ω @ 28 Vdc; 12.8 Ω @ 32 Vdc;
- maximum load inductance: 1000 mH @ 2.5 A;
- for inductive loads with characteristics out of these requirements a dedicated evaluation shall be done;
- current sense accuracy:
  - without calibration procedure (default): ±35% in range (0.5÷1) A, ±30% in range (1÷2) A, ±20% in range (2÷2.5) A;
  - with calibration procedure (custom solution[10]): ±13% in range (0.5÷2.5) A.

Maximum current consumption for ECU is 32 A.
ECU is not self-protected against total overcurrent (>32 A).

NOTE: all outputs are deactivated via software, over 16 Vdc (12 Vdc application) or over 32 Vdc (24 Vdc application).

H-BRIDGE OUTPUTS

Actua has one output to drive a DC electrical motor or other resistive loads only for 12 Vdc applications.

H-Bridge output load characteristics

Resistive loads:

- maximum continuous current = 3 A.

DC electrical motor loads:

- maximum continuous current = 3 A;
- maximum peak current = 15 A[11].

NOTE: all outputs are deactivated via software, over 16 Vdc (12 Vdc application). H-Bridge outputs are not available for 24 Vdc applications.

5. Unreliable measure for input values less than 3 Vdc.
6. Unreliable measure for input values less than 3 Vdc.
7. Unreliable measure for input values less than 30 Ω.
8. See pinout table for output driven in PWM mode.
9. A dedicated calibration procedure can be performed by the final customer on each individual device to increase the accuracy value; MTA may evaluate and quote the development of a dedicated calibration procedure, according to customer specifications.
10. A dedicated calibration procedure can be performed by the final customer on each individual device to increase the accuracy value; MTA may evaluate and quote the development of a dedicated calibration procedure, according to customer specifications.
11. Peak current can be applied for 1 sec maximum.

AUXILIARY SUPPLIES

Two auxiliary supplies for external sensors are configurable at 5 Vdc or 8 Vdc (accuracy: ±4% FS) completely independent, with 200 mA maximum current for each one.
An additional 15 Vdc (accuracy: ±4% FS) sensor supply with 100 mA current capability is available for 24 Vdc systems only, in a range of 18÷32 Vdc[12] (optional).

NOTE: the first two auxiliary outputs work correctly if battery voltage is more than 6 Vdc.

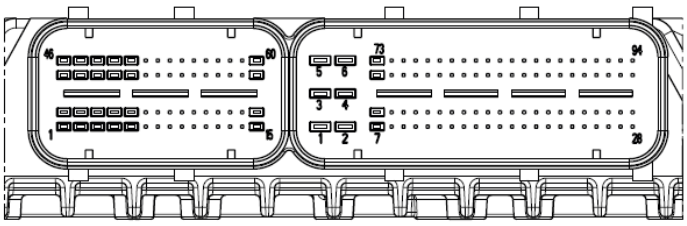
LOAD DUMP

Actua is not provided with load dump protection for 24 Vdc systems but only for 12 Vdc systems.

- load dump test for 12 Vdc system:
  Ua = 14 Vdc ± 0.2 Vdc; Us = 101 Vdc; Ri = 4 Ω; Td = 400 mS; Tr = 0 mS.
  Any other type of load dump, different from tests specified above, has to be evaluated.
- load dump test for 24 Vdc system: not applicable. No protection provided.

CONNECTOR

Tyco 284617.
Product name: 60 way (B pocket) + 94 way (A pocket) pin header.



Features:

- product name: 154 w pin/tab header;
- mating force: ≤ 75 N;
- temperature range: (-40 to +100 °C);
- tab: 2.8 (6 pin), 1.5 (28 pin), 0.63 (120 pin).

PINOUTS

PIN	N°	NAME	TYPE – ECU ACTUA FULL	TYPE – ECU ACTUA OTS1
A01	1	GND	Terminal 31 – Battery ground	Terminal 31 – Battery ground
A02	2	GND	Terminal 31 – Battery ground	Terminal 31 – Battery ground
A03	3	VB	Terminal 30 – Battery power supply	Terminal 30 – Battery power supply
A04	4	VB	Terminal 30 – Battery power supply	Terminal 30 – Battery power supply
A05	5	VB	Terminal 30 – Battery power supply	Terminal 30 – Battery power supply
A06	6	VB	Terminal 30 – Battery power supply	Terminal 30 – Battery power supply
A07	7	OUT_13	High side / low side output with 2.5 A maximum current	High side / low side output with 2.5 A maximum current
A08	8	IN_37	Input configurable as analog or digital	Input configurable as analog or digital
A09	9	IN_28	Input configurable as analog or digital	Input configurable as analog or digital
A10	10	IN_30	Input configurable as analog or digital	Input configurable as analog or digital
A11	11	GND	Terminal 31 – Battery ground	Terminal 31 – Battery ground
A12	12	IN_17	Input configurable as analog or digital	Input configurable as analog or digital
A13	13	IN_19	Input configurable as analog or digital	Input configurable as analog or digital
A14	14	IN_15	Input configurable as analog or digital	Input configurable as analog or digital
A15	15	IN_14	Input configurable as analog or digital	Input configurable as analog or digital
A16	16	GND	Terminal 31 – Battery ground	Terminal 31 – Battery ground
A17	17	IN_5	Input configurable as analog or digital	Input configurable as analog or digital
A18	18	IN_3	Input configurable as analog or digital	Input configurable as analog or digital
A19	19	IN_1	Input configurable as analog or digital	Input configurable as analog or digital
A20	20	VCTRL	Terminal 15 – Ignition input	Terminal 15 – Ignition input
A21	21	GND	Terminal 31 – Battery ground	Terminal 31 – Battery ground
A22	22	USB1_D+	USB Plus	Reserved
A23	23	IN_45	Input configurable as analog or digital (redundant)	Input configurable as analog or digital (redundant)
A24	24	IN_46	Input configurable as analog or digital (redundant)	Input configurable as analog or digital (redundant)
A25	25	IN_47	Input configurable as analog or digital (redundant)	Input configurable as analog or digital (redundant)
A26	26	IN_48	Input configurable as analog or digital (redundant)	Input configurable as analog or digital (redundant)
A27	27	GND	BroadR-Reach shield	Reserved
A28	28	ETH1_BR+	BroadR-Reach plus	Reserved
A29	29	OUT_14	High side / low side output with 2.5 A maximum current	High side / low side output with 2.5 A maximum current
A30	30	IN_23	Input configurable as analog or digital	Input configurable as analog or digital
A31	31	IN_24	Input configurable as analog or digital	Input configurable as analog or digital
A32	32	IN_32	Input configurable as analog or digital	Input configurable as analog or digital

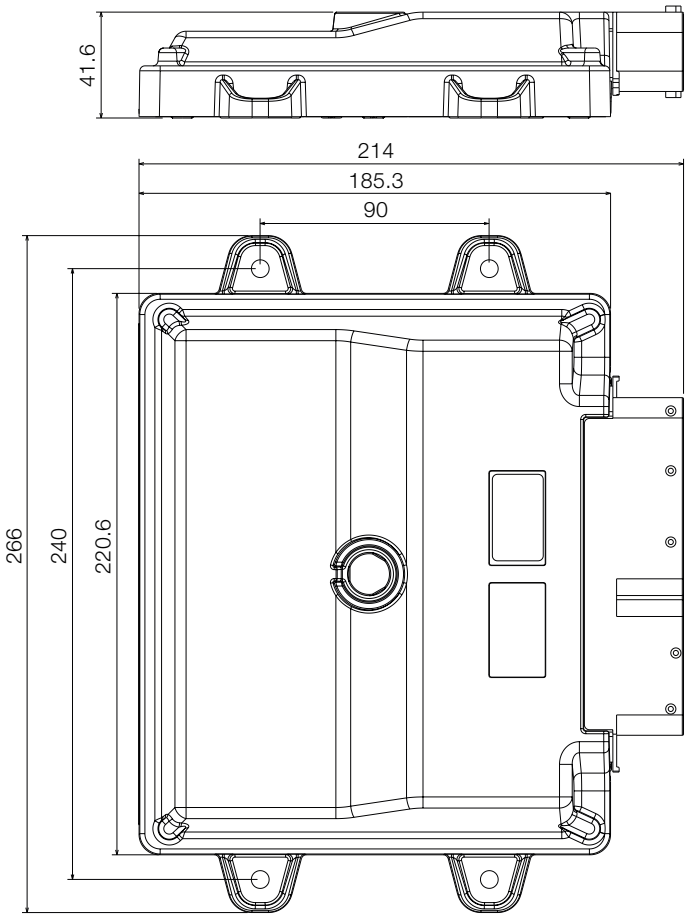
12. Under 18 Vdc, output voltage of 15 Vdc is not guaranteed.

PIN	N°	NAME	TYPE – ECU ACTUA FULL	TYPE – ECU ACTUA OTS1
A33	33	GND	Terminal 31 – Battery ground	Terminal 31 – Battery ground
A34	34	IN_35	Input configurable as analog or digital	Input configurable as analog or digital
A35	35	IN_18	Input configurable as analog or digital	Input configurable as analog or digital
A36	36	IN_11	Input configurable as analog or digital	Input configurable as analog or digital
A37	37	IN_16	Input configurable as analog or digital	Input configurable as analog or digital
A38	38	GND	Terminal 31 – Battery ground	Terminal 31 – Battery ground
A39	39	IN_6	Input configurable as analog or digital	Input configurable as analog or digital
A40	40	IN_4	Input configurable as analog or digital	Input configurable as analog or digital
A41	41	IN_2	Input configurable as analog or digital	Input configurable as analog or digital
A42	42	GND	Terminal 31 – Battery ground	Terminal 31 – Battery ground
A43	43	GND	Terminal 31 – Battery ground	Terminal 31 – Battery ground
A44	44	USB1_D-	USB minus	Reserved
A45	45	IN_43	Input configurable as analog or digital (redundant)	Input configurable as analog or digital (redundant)
A46	46	IN_44	Input configurable as analog or digital (redundant)	Input configurable as analog or digital (redundant)
A47	47	UART1_TX	RS-232	Reserved
A48	48	UART1_RX	RS-232	Reserved
A49	49	GND	RS-232 GND	Reserved
A50	50	ETH1_BR-	BroadR-Reach minus	Reserved
A51	51	OUT_15	High side / low side output with 2.5 A maximum current	High side / low side output with 2.5 A maximum current
A52	52	IN_25	Input configurable as analog or digital	Input configurable as analog or digital
A53	53	IN_27	Input configurable as analog or digital	Input configurable as analog or digital
A54	54	IN_31	Input configurable as analog or digital	Input configurable as analog or digital
A55	55	IN_34	Input configurable as analog or digital	Input configurable as analog or digital
A56	56	IN_20	Input configurable as analog or digital	Input configurable as analog or digital
A57	57	IN_9	Input configurable as analog or digital	Input configurable as analog or digital
A58	58	IN_12	Input configurable as analog or digital	Input configurable as analog or digital
A59	59	GND	Terminal 31 – Battery ground	Terminal 31 – Battery ground
A60	60	IN_21	Input configurable as analog or digital	Input configurable as analog or digital
A61	61	IN_22	Input configurable as analog or digital	Input configurable as analog or digital
A62	62	CAN1_H	CAN bus (CAN1 pass through)	CAN bus (CAN1 pass through)
A63	63	CAN1_L	CAN bus (CAN1 pass through)	CAN bus (CAN1 pass through)
A64	64	GND	CAN1 shield	CAN1 shield
A65	65	CAN2_L	CAN bus (CAN2 pass through)	CAN bus (CAN2 pass through)
A66	66	CAN2_H	CAN bus (CAN2 pass through)	CAN bus (CAN2 pass through)
A67	67	IN_41	Input configurable as analog or digital (redundant)	Input configurable as analog or digital (redundant)
A68	68	CAN3_L	CAN bus (CAN3 pass through)	CAN bus (CAN3 pass through)
A69	69	CAN3_H	CAN bus (CAN3 pass through)	CAN bus (CAN3 pass through)
A70	70	IN_42	Input configurable as analog or digital (redundant)	Input configurable as analog or digital (redundant)
A71	71	CAN4_L	CAN bus (CAN4 pass through)	CAN bus (CAN4 pass through)
A72	72	CAN4_H	CAN bus (CAN4 pass through)	CAN bus (CAN4 pass through)
A73	73	OUT_16	High side / low side output with 2.5 A maximum current	High side / low side output with 2.5 A maximum current
A74	74	GND	Terminal 31 – Battery ground	Terminal 31 – Battery ground
A75	75	IN_26	Input configurable as analog or digital	Input configurable as analog or digital
A76	76	IN_29	Input configurable as analog or digital	Input configurable as analog or digital
A77	77	IN_33	Input configurable as analog or digital	Input configurable as analog or digital
A78	78	IN_36	Input configurable as analog or digital	Input configurable as analog or digital
A79	79	IN_10	Input configurable as analog or digital	Input configurable as analog or digital
A80	80	IN_13	Input configurable as analog or digital	Input configurable as analog or digital
A81	81	GND	CAN1 Shield	CAN1 shield
A82	82	LIN1	LIN bus	LIN bus

PIN	N°	NAME	TYPE – ECU ACTUA FULL	TYPE – ECU ACTUA OTS1
A83	83	LIN2	LIN bus	LIN bus
A84	84	CAN1_H	CAN bus	CAN bus
A85	85	CAN1_L	CAN bus	CAN bus
A86	86	GND	CAN2 shield	CAN2 shield
A87	87	CAN2_L	CAN bus	CAN bus
A88	88	CAN2_H	CAN bus	CAN bus
A89	89	GND	CAN3 shield	CAN3 shield
A90	90	CAN3_L	CAN bus	CAN bus
A91	91	CAN3_H	CAN bus	CAN bus
A92	92	GND	CAN4 shield	CAN4 shield
A93	93	CAN4_L	CAN bus	CAN bus
A94	94	CAN4_H	CAN bus	CAN bus
B01	95	OUT_1	High side output with 4 A maximum current and current sense	High side output with 4 A maximum current and current sense
B02	96	OUT_18	High side output with 2.5 A maximum current (no PWM)	High side output with 2.5 A maximum current (no PWM)
B03	97	OUT_5	High side output with 2.5 A maximum current and current sense	High side output with 2.5 A maximum current and current sense
B04	98	OUT_6	High side output with 2.5 A maximum current and current sense	High side output with 2.5 A maximum current and current sense
B05	99	OUT_7	High side output with 2.5 A maximum current and current sense	High side output with 2.5 A maximum current and current sense
B06	100	GND	Terminal 31 – Battery ground	Terminal 31 – Battery ground
B07	101	IN_49	Input configurable as frequency or digital	Input configurable as frequency or digital
B08	102	IN_50	Input configurable as frequency or digital	Input configurable as frequency or digital
B09	103	IN_51	Input configurable as frequency or digital	Input configurable as frequency or digital
B10	104	IN_52	Input configurable as frequency or digital	Input configurable as frequency or digital
B11	105	GND	Terminal 31 – Battery ground	Terminal 31 – Battery ground
B12	106	IN_53	Input configurable as frequency or digital	Input configurable as frequency or digital
B13	107	IN_54	Input configurable as frequency or digital	Input configurable as frequency or digital
B14	108	IN_55	Input configurable as frequency or digital	Input configurable as frequency or digital
B15	109	OUT_12	High side output with 2.5 A maximum current and current sense	High side output with 2.5 A maximum current and current sense
B16	110	OUT_2	High side output with 4 A maximum current and current sense	High side output with 4 A maximum current and current sense
B17	111	OUT_25	High side output with 2.5 A maximum current (no PWM)	High side output with 2.5 A maximum current (no PWM)
B18	112	OUT_26	High side output with 2.5 A maximum current (no PWM)	High side output with 2.5 A maximum current (no PWM)
B19	113	OUT_17	High side output with 2.5 A maximum current (no PWM)	High side output with 2.5 A maximum current (no PWM)
B20	114	OUT_8	High side output with 2.5 A maximum current and current sense	High side output with 2.5 A maximum current and current sense
B21	115	IN_56	Input configurable as frequency or digital	Input configurable as frequency or digital
B22	116	GND	Terminal 31 – Battery ground	Terminal 31 – Battery ground
B23	117	GND	Terminal 31 – Battery ground	Terminal 31 – Battery ground
B24	118	IN_8	Input configurable as analog or digital	Input configurable as analog or digital
B25	119	IN_7	Input configurable as analog or digital	Input configurable as analog or digital
B26	120	GND	Terminal 31 – Battery ground	Terminal 31 – Battery ground
B27	121	IN_38	Input configurable as analog or digital	Input configurable as analog or digital
B28	122	IN_39	Input configurable as analog or digital	Input configurable as analog or digital
B29	123	GND	Terminal 31 – Battery ground	Terminal 31 – Battery ground
B30	124	OUT_11	High side output with 2.5 A maximum current and current sense	High side output with 2.5 A maximum current and current sense
B31	125	OUT_33	DC motor output	DC motor output
B32	126	OUT_27	High side output with 2.5 A maximum current (no PWM)	High side output with 2.5 A maximum current (no PWM)

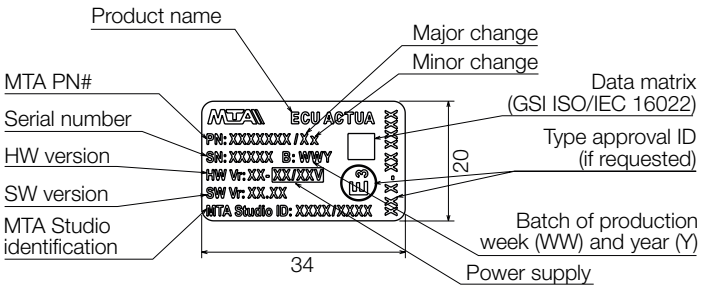
PIN	N°	NAME	TYPE – ECU ACTUA FULL	TYPE – ECU ACTUA OTS1
B33	127	OUT_24	High side output with 2.5 A maximum current (no PWM)	High side output with 2.5 A maximum current (no PWM)
B34	128	OUT_19	High side output with 2.5 A maximum current (no PWM)	High side output with 2.5 A maximum current (no PWM)
B35	129	OUT_21	High side output with 2.5 A maximum current (no PWM)	High side output with 2.5 A maximum current (no PWM)
B36	130	OUT_23	High side output with 2.5 A maximum current (no PWM)	High side output with 2.5 A maximum current (no PWM)
B37	131	GND	Terminal 31 – Battery ground	Terminal 31 – Battery ground
B38	132	IN_57	Analog input	Analog input
B39	133	IN_58	Analog input	Analog input
B40	134	IN_40	Input configurable as analog or digital	Input configurable as analog or digital
B41	135	GND	Terminal 31 – Battery ground	Terminal 31 – Battery ground
B42	136	VSS_1	Configurable auxiliary supply 5 Vdc or 8 Vdc	Configurable auxiliary supply 5 Vdc or 8 Vdc
B43	137	VSS_3	15 Vdc sensor supply – 24 Vdc systems only	Reserved
B44	138	VSS_2	Configurable auxiliary supply 5 Vdc or 8 Vdc	Configurable auxiliary supply 5 Vdc or 8 Vdc
B45	139	OUT_4	High side output with 4 A maximum current and current sense	High side output with 4 A maximum current and current sense
B46	140	OUT_34	DC motor output	DC motor output
B47	141	OUT_28	High side output with 2.5 A maximum current (no PWM)	High side output with 2.5 A maximum current (no PWM)
B48	142	OUT_20	High side output with 2.5 A maximum current (no PWM)	High side output with 2.5 A maximum current (no PWM)
B49	143	OUT_30	High side output with 2.5 A maximum current (no PWM)	High side output with 2.5 A maximum current (no PWM)
B50	144	OUT_22	High side output with 2.5 A maximum current (no PWM)	High side output with 2.5 A maximum current (no PWM)
B51	145	OUT_29	High side output with 2.5 A maximum current (no PWM)	High side output with 2.5 A maximum current (no PWM)
B52	146	OUT_9	High side output with 2.5 A maximum current and current sense	High side output with 2.5 A maximum current and current sense
B53	147	OUT_10	High side output with 2.5 A maximum current and current sense	High side output with 2.5 A maximum current and current sense
B54	148	OUT_32	High side output with 2.5 A maximum current (no PWM)	High side output with 2.5 A maximum current (no PWM)
B55	149	OUT_31	High side output with 2.5 A maximum current (no PWM)	High side output with 2.5 A maximum current (no PWM)
B56	150	GND	Terminal 31 – Battery ground	Terminal 31 – Battery ground
B57	151	VSS_1	Configurable auxiliary supply 5 Vdc or 8 Vdc	Configurable auxiliary supply 5 Vdc or 8 Vdc
B58	152	VSS_3	15 Vdc sensor supply – 24 Vdc systems only	Reserved
B59	153	VSS_2	Configurable auxiliary supply 5 Vdc or 8 Vdc	Configurable auxiliary supply 5 Vdc or 8 Vdc
B60	154	OUT_3	High side output with 4 A maximum current and current sense	High side output with 4 A maximum current and current sense

DIMENSIONS



All the PN# of ECU Actua shares the same mechanical case.

LABEL DETAILS



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Please check on MTA website the latest version of this technical data sheet.

MATING CONNECTOR

Mating connector used for wiring harness is a Tyco 94 + 60 pins.

The following parts are required for assembling a wiring harness connector:
- 94-pin connector PN# 284743-1 (qty: 1);
- 60-pin connector PN# 284742-1 (qty: 1).

NAME	CRIMP SECTION [mm²]	PART NUMBER
MQS clean body	0.2 ÷ 0.5	968220-1
MQS clean body	0.75	968221-1
MQS 1.5 clean body	0.5	1452158-1
MQS 1.5 clean body	0.75 ÷ 1.5	1241608-1
MCP 2.8 K	0.5 ÷ 1.0	1241394-1
MCP 2.8 K	>1.0 ÷ 2.5	1241396-1
A33	GND	GND for inputs
A38	GND	GND for inputs
A42	GND	GND for inputs
A43	GND	GND for inputs
A59	GND	GND for inputs
A74	GND	GND for inputs
A81	GND	GND for inputs
B26	GND	GND for inputs
B29	GND	GND for inputs
B37	GND	GND for inputs

SAFETY FEATURES

Actua contains following features suitable for the functional safety:
- watchdog management with Q&A protocol via SPI (microcontroller & System Basis Chip);
- hardware circuitry that guarantees output switch-off when the microcontroller is in “reset” state (default condition).

NOTE: details of each safety feature listed above depend from the vehicle where the ECU is mounted and from the functional safety requested by the customer.

PRODUCT QUALIFICATION

The MTA reference test specification for ECU Actua is TS50_16 (rev.6) which includes:
- environmental tests;
- mechanical tests;
- chemical tests;
- electrical tests;
- EMC tests;
- CE type approval.

Please contact MTA for any detail.

NOTE: ECU Actua Full (PN# 6200156/6x) is still not qualified; check with MTA the availability of this PN# that includes the following optional contents:
1. USB: only device mode available for service purposes;
2. Ethernet BroadR-Reach;
3. RS-232 not available as vehicle communication bus;
4. accelerometer;
5. serial flash memory;
6. auxiliary power supplies (15 Vdc for 24 Vdc applications).

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